

BIO102 HELP SESSION 3

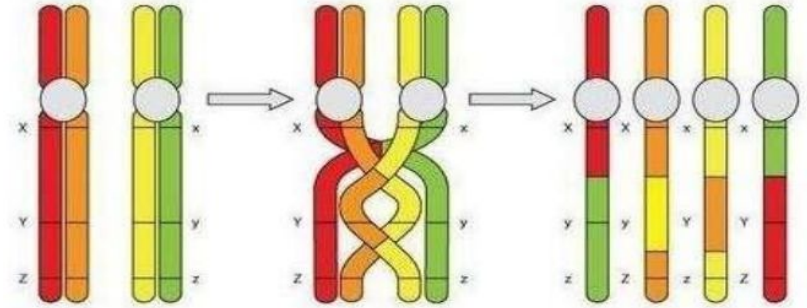
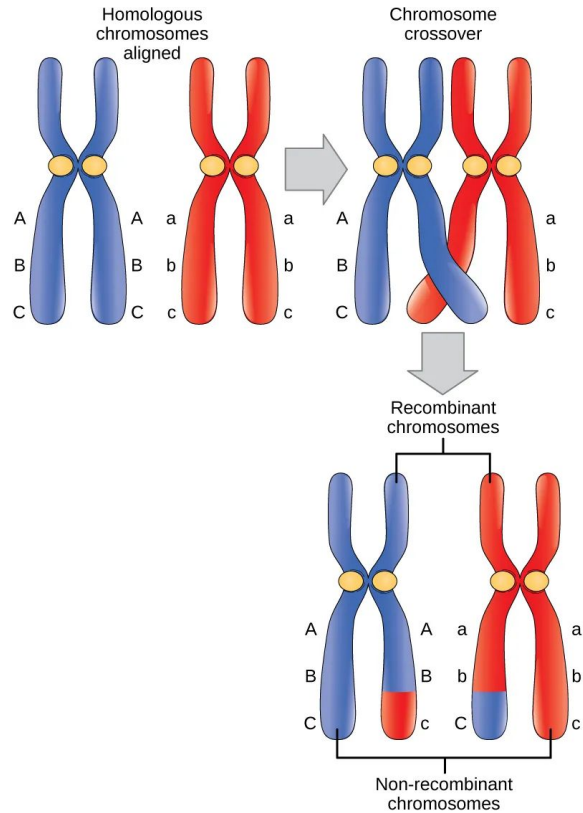
Genetic linkage

Genetic Linkage: It is the tendency of alleles at different loci to be inherited together, more often than predicted by chance alone. This usually happens because the two loci are very close together on the same chromosome. Such loci are called Linked Loci.

LINKAGE AFFECTS INDEPENDENT ASSORTMENT OF ALLELES AT DIFFERENT LOCI

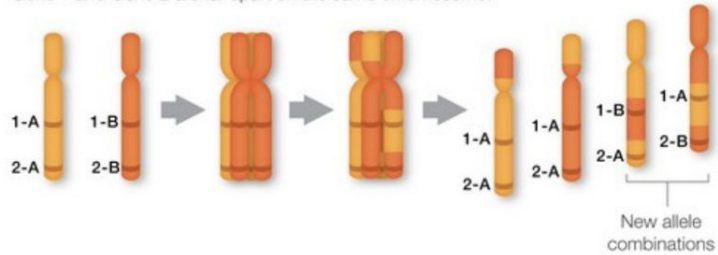
This leads to an over-representation of parental allele combinations and an under representation of recombinant allele combinations in the gamete pool.

Crossing Over



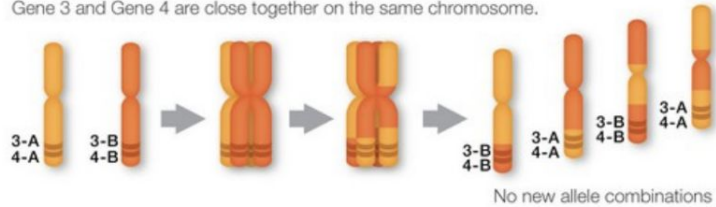
Not Linked

Gene 1 and Gene 2 are far apart on the same chromosome.



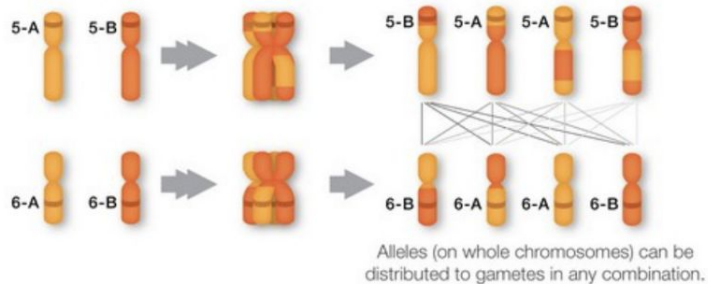
Linked

Gene 3 and Gene 4 are close together on the same chromosome.



Not Linked

Gene 5 and Gene 6 are on separate chromosomes.



5.1 What is the principal difference between the genotype $A B / a b$ and the genotype $A b / a B$? What is the role of the slash (/, technically called a virgule) in such an expression? Which genotype is the coupling (*cis*) genotype and which is the repulsion (*trans*) genotype?

5.1 What is the principal difference between the genotype $A B / a b$ and the genotype $A b / a B$? What is the role of the slash (/, technically called a virgule) in such an expression? Which genotype is the coupling (*cis*) genotype and which is the repulsion (*trans*) genotype?

ANS 5.1. In the genotype $A B / a b$, the two wildtype alleles are in the same chromosome, in the genotype $A b / a B$, they are in different chromosomes. The slash separates the alleles present in one chromosome from those present in the homologous chromosome. The $A B / a b$ genotype is in coupling (*cis*), the $A b / a B$ genotype in repulsion (*trans*).

5.2 What gametes, and in what ratios, are produced by male and female *Drosophila* of genotype $A B a b$ when the genes are present in the same chromosome and the frequency of recombination between them is 5 percent?

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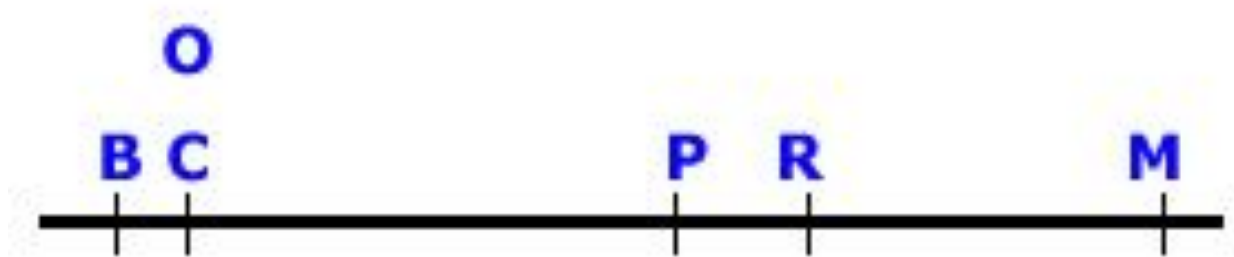
ANS 5.2. There is no crossing-over in male *Drosophila*, and so the gametes will be $A B$ and $a b$ in equal proportions. In the female, each of the nonrecombinant gametes $A B$ and $a b$ is expected with a frequency of $(1 - 0.05)/2 = 0.475$, and each of the recombinant gametes $A b$ and $a B$ is expected with a frequency of $0.05/2 = 0.025$.

Linkage maps

Linkage Map:

This is a map of the loci on chromosomes. The distances between loci indicate the frequency of recombination between them. Hence, this is also called “linkage map”. Please note that this is not a physical map of the chromosome.

The Map Unit is centiMorgans. According to A H Sturtevant, 1% recombination is equal to one centiMorgan (cM).



5.5 If the recombination frequency between genes *A* and *B* is 6.2 percent, what is the distance between the genes in map units in the linkage map?

Linkage Maps

if the frequency of crossing over was related to distance, one could use this information to map out the genes on a chromosome.

After all, the farther apart two genes were on a chromosome, the more likely it was that these genes would separate during recombination. Therefore, as Sturtevant explained it, the **"proportion of crossovers could be used as an index of the distance between any two factors"**

$$\text{Crossing Over Frequency (\%)} = \frac{\text{number of recombinants}}{\text{Total offspring}} \times 100$$

2. Crossover map distances determined by two-point crosses are $P-C = 7$, $S-M = 10$, $C-M = 8$, $S-C = 2$, and $P-S = 5$. Draw a map of these four linked loci.

Two-Point Test Cross

F₁

Red eye, Normal wing X purple eye, Vestigial wing



F ₁ Gamete	Testcross Distribution	Gamete Type
<i>pr⁺ vg⁺</i>	1339	Parental
<i>pr⁺ vg</i>	151	Recombinant
<i>pr vg⁺</i>	154	Recombinant
<i>pr vg</i>	1195	Parental
Total	2839	

Frequency of Recombinants = $(151 + 154) / 2839 = 0.107$



Three Point Test Cross

Genotype	Observed	Type of Gamete
$v\ cv^+\ ct^+$	580	Parental
$v^+\ cv\ ct$	592	Parental
$v\ cv\ ct^+$	45	Single-crossover between genes ct and cv
$v^+\ cv^+\ ct$	40	Single-crossover between genes ct and cv
$v\ cv\ ct$	89	Single-crossover between genes v and ct
$v^+\ cv^+\ ct^+$	94	Single-crossover between genes v and ct
$v\ cv^+\ ct$	3	Double-crossover
$v^+\ cv\ ct^+$	5	Double-crossover
Total	1448	

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Step 1: determine the Parental and Recombinant types

- Parentals
- Single Recombinants (Single Cross Overs)
- Double Recombinants (Double Cross Overs)

Step 2: Determine the Gene Order.

Step 3: Determine the linkage distances.

Step 4: Draw the map.

Step 5: Calculate Interference.

Interference (I) = 1- Coefficient of Coincidence

C.o.C = $\frac{\text{Observed number of double recombinants}}{\text{Expected number of double recombinants}}$

Three Point Test Cross - Example

Genotype	Observed	Type of Gamete
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Step 1: Determine the parental genotypes.

The most abundant genotypes are the parental types. These genotypes are $v\ cv^+\ ct^+$ and $v^+\ cv\ ct$.

Step 2: Determine the gene order

To determine the gene order, we need the parental genotypes as well as the double crossover genotypes. The least frequent genotypes are the double-crossover genotypes. These genotypes are $v\ cv^+\ ct$ and $v^+\ cv\ ct^+$

In the double-crossover genotypes, which parental allele is not associated with the two parental alleles it was associated with in the original parental cross?

From the first double crossover, $v\ cv^+\ ct$, the ct allele is associated with the v and cv^+ alleles, two alleles it was not associated with in the original cross. Therefore, ct is in the middle, and the gene order is $v\ ct\ cv$.

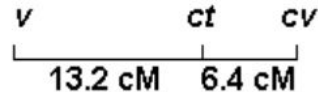
Three Point Test Cross - Example

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$v^+\ cv\ ct^+$	5	Double-crossover
Total	1448	

Step 3: Determining the linkage distances.

- v - ct distance calculation. This distance is derived as follows:
 $100 * ((89+94+3+5)/1448) = 13.2\text{ cM}$
- ct - cv distance calculation. This distance is derived as follows:
 $100 * ((45+40+3+5)/1448) = 6.4\text{ cM}$

Step 4. Draw the map.



Three Point Test Cross - Example

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Total	1448	

Step 5: Calculate Interference

The amount of double crossover gives an indication if interference occurs. Given specific recombination rates in two adjacent chromosomal intervals, the rate of double-crossovers in this region should be equal to the product of the single crossovers. I

Recombination frequency is 0.132 between genes v and ct , and the recombination frequency between ct and cv is 0.064.

Therefore, we would expect 0.84% [$100 \times (0.132 \times 0.064)$] double recombinants. With a sample size of 1448, this would amount to 12 double recombinants. We actually only detected 8.

To measure interference, we first calculate the **coefficient of coincidence (c.o.c.)** which is the ratio of observed to expected double recombinants. Interference is then calculated as $1 - \text{c.o.c.}$. The formula is as follows:

$$I = 1 - \text{c.o.c.} = 1 - \left[\frac{\text{Observed \# of double recombinants}}{\text{Expected \# of double recombinants}} \right]$$

The interference value is 33% [$100 \times (8/12)$].

Most often, interference values fall between 0 and 1. Values less than one indicate that interference is occurring in this region of the chromosome.

TABLE 9.3. Summary of the results obtained from a three point test cross between $ABC/abc \times abc/abc$

<i>Genotypic classes</i>	<i>Phenotypic classes</i>	<i>Assumed frequencies</i>	<i>Remarks</i>
ABC/abc	ABC	349	} Parental types.
abc/abc	abc	360	
Abc/abc	Abc	114	} Single crossover between A and B.
aBC/abc	aBC	116	
ABc/abc	ABc	128	} Single crossover between B and C.
abC/abc	abC	124	
AbC/abc	AbC	5	} Double crossover between A and C.
aBc/abc	aBc	4	
Total		1200	

2. Recombination (%) of Three point Test Cross:

Crossing over percentage between genes A and B = $(114+116)+(5+4)/1200 \times 100 = 230 + 9/1200 \times 100 = 19.92\%$

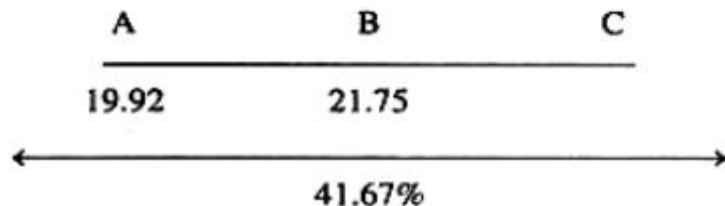
Crossing over percentage between genes B and C = $(128 + 124)+ (5 + 4)/1200 \times 100 = 252 + 9/1200 \times 100 = 21.75\%$

3. Gene Sequence of Three Point Test Cross:

Hence, distance between genes A & B = 19.92 map units

Distance between genes B& C= 21.75 map units

Distance between genes A & C= $21.75 + 19.92 = 41.67$ map units



4. Coefficient of Coincidence in Three point Test Cross:

Coefficient of coincidence =

Observed no. of double crossovers/Expected no. of double crossovers

1. Observed double crossovers = $9/1200 \times 100 = 0.75$

2. Expected double crossovers = Product of two single crossovers between AB & BC values/100

= $19.92 \times 21.75/100 = 4.33$

Coefficient of coincidence = $0.75/4.33 = 0.17$

Coefficient of interference = $1 - \text{Coefficient of coincidence} = 1 - 0.17 = 0.83$

Question 1

You are doing a genetics experiment with the fruit fly. In the “P” generation, you cross two true-breeding flies. The female parent is brown and wingless and the male parent is black with normal wings. All of the flies in the F1 generation are brown and have normal wings.

Indicate the alleles associated with dominant phenotypes by uppercase letters and alleles associated with recessive phenotypes by lowercase letters. Assume the genes are not found on a sex chromosome. Indicate the color alleles as “B” and “b” the wing alleles by the letters “N” and “n”.

a) The genotypes of the flies in the P generation are:

_____ female and _____ male.

b) The genotype of the flies in the F1 generation is: _____

c) You now take an F1 female and cross her to a true-breeding black, wingless male.

This male's genotype is: _____ .

a) The genotypes of the flies in the P generation are:

BBnn female and **bbNN** male.

b) The genotype of the flies in the F1 generation is: **BbNn**

c) You now take an F1 female and cross her to a true-breeding black, wingless male.

This male's genotype is: **bbnn**

d) You count 1600 offspring in the F2 generation. If the wing and the color traits were linked and no recombination occurred, you would expect to count:

_____ # of brown, winged flies (of the genotype _____)

_____ # of black, winged flies (of the genotype _____)

_____ # of brown, wingless flies (of the genotype _____)

_____ # of black, wingless flies (of the genotype _____)

e) When you count the F2 generation, you really get:

- 85 brown winged flies
- 728 black winged flies
- 712 brown wingless flies
- 75 black wingless flies

What is the genetic distance between the color and wing genes?

0# of brown, winged flies (of the genotype BbNn)

800# of black, winged flies (of the genotype Bbnn)

800# of brown, wingless flies (of the genotype bbNn)

0# of black, wingless flies (of the genotype bbnn)

e) When you count the F2 generation, you really get:

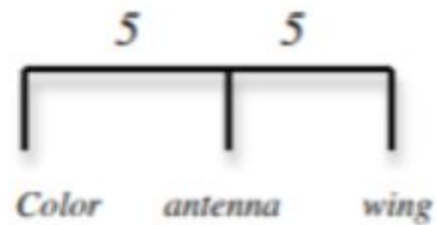
- 85 brown winged flies
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What is the genetic distance between the color and wing genes?

Recombination frequency = $\frac{\# \text{ recombinants}}{\text{total}} (100) = \frac{85 + 75}{85 + 75 + 728 + 712} = 0.1(100) = 10\%$

f) A series of fruit fly matings shows that the recombination frequency between the gene for wing size and the gene for antenna length is 5% (i.e. the genetic distance between them is 5 centimorgans). List all possible recombination frequencies between the gene for color and the gene for antenna length.

The genetic distance is either 5 cM or 15 cM



Question 2

You are working with a hypothetical fly and have found color and wing mutants. Preliminary work indicates that the mutant traits are recessive and the associated genes are not sex-linked, but beyond that you have no information. You first look at 2 genes, each with two alleles: "B or b" for body color and "W or w" for wing surface. The red body phenotype is dominant to the yellow body phenotype and smooth wings are dominant to wrinkled wings.

a) You cross a true-breeding yellow-bodied, smooth-winged female with a true-breeding red-bodied, crinkle winged male. What will be the phenotype(s) of the F1 progeny?

b) You cross several pairs of F1 siblings.

i) What ratio of phenotypes do you expect in the offspring?

red smooth: red crinkled: yellow smooth: yellow crinkled

a) You cross a true-breeding yellow-bodied, smooth-winged female with a true-breeding red-bodied, crinkle winged male. What will be the phenotype(s) of the F1 progeny?

$RRaa \times rrAA = RrAa$ Thus the F1 fly has a red body and smooth wings.

b) You cross several pairs of F1 siblings.

i) What ratio of phenotypes do you expect in the offspring?

9 : 3 : 3 : 1

red smooth: red crinkled: yellow smooth: yellow crinkled

ii) The actually data is shown below. Does this date match your expectation? If not, explain why there is a difference in the expected and actual numbers.

body	wing surface	Number
red	smooth	310
yellow	smooth	142
red	crinkled	131
yellow	crinkled	23

c) To determine the recombination frequency between these two genes, you perform several crosses where you cross an F1 from part (a) with a yellow-bodied, crinkle-winged fly. You get the following results. What is the distance between the genes for body color and wing surface in map units?

body	wing surface	Number
red	crinkled	396
red	smooth	102
yellow	crinkled	98
yellow	smooth	404

It is hard to tell, although this ratio does not match the 9:3:3:1 expected, but it is not terribly far off. You would suspect that these two genes may be linked, but doing a test cross (back cross) will give much clearer results.

c) To determine the recombination frequency between these two genes, you perform several crosses where you cross an F1 from part (a) with a yellow-bodied, crinkle-winged fly. You get the following results. What is the distance between the genes for body color and wing surface in map units?

body	wing surface	Number
red	crinkled	396
red	smooth	102
yellow	crinkled	98
yellow	smooth	404

$$(102+98)/(396+102+98+404) * 100 = 20 \text{ cM}$$

You decide to turn your attention to a different gene, one that controls wing length. This gene has two alleles, "L or l" where long wings are dominant to short wings. Remember that the red body phenotype is dominant to the yellow body phenotype. You again mate two true-breeding lines:

red-bodied, short wing male X yellow-bodied, long wing female

F1: All red-bodied, long wing

d) To determine the recombination frequency between these two genes, you perform several crosses where you cross an F1 from part (d) with a yellow-bodied, short-winged fly. You get the following results. What is the distance between the genes for body color and wing length in map units?

body	wing length	Number
red	long	45
red	short	460
yellow	long	440
yellow	short	55

e) Given your answers to parts (c) and (d), what can you say about the relationship between the gene for wing length and the gene for wing surface?

d) To determine the recombination frequency between these two genes, you perform several crosses where you cross an F1 from part (d) with a yellow-bodied, short-winged fly. You get the following results. What is the distance between the genes for body color and wing length in map units?

body	wing length	Number
red	long	45
red	short	460
yellow	long	440
yellow	short	55

$$(45+55)/(45+460+440+55) * 100 = 10 \text{ cM}$$

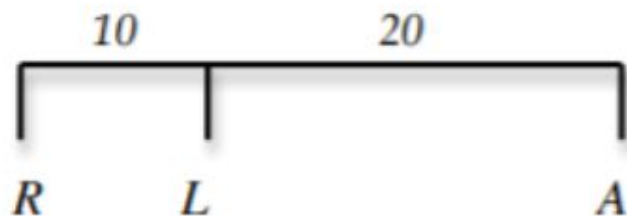
e) Given your answers to parts (c) and (d), what can you say about the relationship between the gene for wing length and the gene for wing surface?

You would expect the gene for wing length and the gene for wing surface to be linked and the distance is expected to be either approximately 10 cM or approximately 30 cM apart.

f) Based on the preceding, what are the two possible arrangements of the three genes? Indicate the map distance between genes for each arrangement.

g) In a final mapping experiment, you cross a true-breeding red-bodied, short and crinkle-winged male with a true breeding yellow-bodied, long and smooth-winged female. List the expected genotypes of the F1 progeny. What are the associated phenotypes?

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g) In a final mapping experiment, you cross a true-breeding red-bodied, short and crinkle-winged male with a true breeding yellow-bodied, long and smooth-winged female. List the expected genotypes of the F1 progeny. What are the associated phenotypes?

P: $RRllaa \times rrLLAA$

F1: $RrLlAa$ red-bodied, long and smooth-winged

Question 3

While working with a type of beetle that is normally smooth, large, and white, you discover three mutations that lead to the recessive phenotypes bumpy, small, and grey, respectively. You cross true-breeding smooth white beetles to true-breeding bumpy grey beetles and get all smooth white F1 beetles. Then you cross the F1 beetles to true-breeding bumpy grey beetles and, after analyzing 800 F2s, calculate a map distance of 5 cM between the smoothness and color loci.

a) What are the four phenotypic classes you got in the F2, and about how many of each did you get?

- For the body type (i.e., smooth or bumpy) use B or b to designate the alleles.
- For the color (i.e., white or grey) use G or g to designate the alleles.
- For the size (i.e., large or small) use L or l to designate the alleles.

*In each case, use the uppercase letter for the associated with the dominant phenotype and the lower case letter for the allele associated with the recessive phenotype.

<i>Genotypic class</i>	<i>Phenotypic class</i>	<i>How many of this phenotype</i>
<i>Bbgg</i>	<i>Smooth White</i>	380
<i>bbGg</i>	<i>Bumpy Grey</i>	380
<i>bbGb</i>	<i>Smooth Grey</i>	20
<i>bbgg</i>	<i>Bumpy White</i>	20

b) You cross two true-breeding parents to get all F1 beetles that are large and smooth. You cross the F1 to a true-breeding small, bumpy beetles and get: 228 large and smooth, 19 small and smooth, 16 large and bumpy, 237 small and bumpy. What were the genotypes and phenotypes of the two parental beetles (P generation)?

	<i>genotype</i>	<i>phenotype</i>
<i>Parent 1</i>	<i>LLBB</i>	<i>Large Smooth</i>
<i>Parent 2</i>	<i>llbb</i>	<i>Small bumpy</i>